

STAC67: Regression Analysis

Lecture 7

Thibault Randrianarisoa

September 30, 2026

University of Toronto Scarborough

Reminder: Assignment 1 due Friday

Due Friday 2 October, 23:59, on Crowdmark. Late submissions are not accepted.

Submitting on Crowdmark

1. Each question has its own upload slot: put the pages for each question in its slot.
2. Questions 1 and 7: the knitted R Markdown output, with your code and its output visible. If both are in one file, upload that file to both slots.
3. Questions 2 to 6: typed or handwritten. Scans must be clearly legible.
4. Once everything is uploaded, press **Submit** (uploading alone does not submit).
 - Show your work: an answer without justification receives no credit.

Leave yourself time for scanning and uploading: do not start at 23:45.

Review of Lecture 6

At a new point x_{n+1} , the fitted value $\hat{Y}_{n+1} = \hat{\beta}_0 + \hat{\beta}_1 x_{n+1}$ does two different jobs.

Review of Lecture 6

At a new point x_{n+1} , the fitted value $\hat{Y}_{n+1} = \hat{\beta}_0 + \hat{\beta}_1 x_{n+1}$ does two different jobs.

Estimate the mean $\mu_{n+1} = E(Y \mid X = x_{n+1})$ for which we have a **confidence interval**

$$\hat{Y}_{n+1} \pm t(1 - \alpha/2; n - 2) \hat{\sigma} \sqrt{\frac{1}{n} + \frac{(x_{n+1} - \bar{x})^2}{S_{xx}}}$$

Review of Lecture 6

At a new point x_{n+1} , the fitted value $\hat{Y}_{n+1} = \hat{\beta}_0 + \hat{\beta}_1 x_{n+1}$ does two different jobs.

Estimate the mean $\mu_{n+1} = E(Y \mid X = x_{n+1})$ for which we have a **confidence interval**

$$\hat{Y}_{n+1} \pm t(1 - \alpha/2; n - 2) \hat{\sigma} \sqrt{\frac{1}{n} + \frac{(x_{n+1} - \bar{x})^2}{S_{xx}}}$$

Predict one new observation Y_{n+1} for which we have a **prediction interval**

$$\hat{Y}_{n+1} \pm t(1 - \alpha/2; n - 2) \hat{\sigma} \sqrt{\mathbf{1} + \frac{1}{n} + \frac{(x_{n+1} - \bar{x})^2}{S_{xx}}}$$

$$Y_{n+1} = \mu_{n+1} + \underbrace{\varepsilon_{n+1}}_{\sim \mathcal{N}(0, \sigma^2)}$$

Review of Lecture 6

At a new point x_{n+1} , the fitted value $\hat{Y}_{n+1} = \hat{\beta}_0 + \hat{\beta}_1 x_{n+1}$ does two different jobs.

Estimate the mean $\mu_{n+1} = E(Y | X = x_{n+1})$ for which we have a **confidence interval**

$$\hat{Y}_{n+1} \pm t(1 - \alpha/2; n - 2) \hat{\sigma} \sqrt{\frac{1}{n} + \frac{(x_{n+1} - \bar{x})^2}{S_{xx}}}$$

Predict one new observation Y_{n+1} for which we have a **prediction interval**

$$\hat{Y}_{n+1} \pm t(1 - \alpha/2; n - 2) \hat{\sigma} \sqrt{\mathbf{1} + \frac{1}{n} + \frac{(x_{n+1} - \bar{x})^2}{S_{xx}}}$$

ANOVA: $SST = \overbrace{SSR}^{= \sum e_i^2} + \overbrace{SSE}$, and the degrees of freedom split the same way, $n - 1 = 1 + (n - 2)$.

$$F^* = \frac{MSR}{MSE} \stackrel{H_0}{\sim} F(1, n - 2), \quad F^* = (t^*)^2, \quad R^2 = \frac{SSR}{SST} = r^2$$

F^* small indicates that β_1 might be 0, which indicates that X brings no information on Y

Review of Lecture 6

At a new point x_{n+1} , the fitted value $\hat{Y}_{n+1} = \hat{\beta}_0 + \hat{\beta}_1 x_{n+1}$ does two different jobs.

Estimate the mean $\mu_{n+1} = E(Y \mid X = x_{n+1})$ for which we have a **confidence interval**

$$\hat{Y}_{n+1} \pm t(1 - \alpha/2; n - 2) \hat{\sigma} \sqrt{\frac{1}{n} + \frac{(x_{n+1} - \bar{x})^2}{S_{xx}}}$$

Predict one new observation Y_{n+1} for which we have a **prediction interval**

$$\hat{Y}_{n+1} \pm t(1 - \alpha/2; n - 2) \hat{\sigma} \sqrt{\mathbf{1} + \frac{1}{n} + \frac{(x_{n+1} - \bar{x})^2}{S_{xx}}}$$

ANOVA: $SST = SSR + SSE$, and the degrees of freedom split the same way, $n - 1 = 1 + (n - 2)$.

$$F^* = \frac{MSR}{MSE} \stackrel{H_0}{\sim} F(1, n - 2), \quad F^* = (t^*)^2, \quad R^2 = \frac{SSR}{SST} = r^2$$

Today: all of this on a real dataset and the natural question

Are the assumptions behind these intervals and tests actually true?

Case study: COVID-19 in Chicago

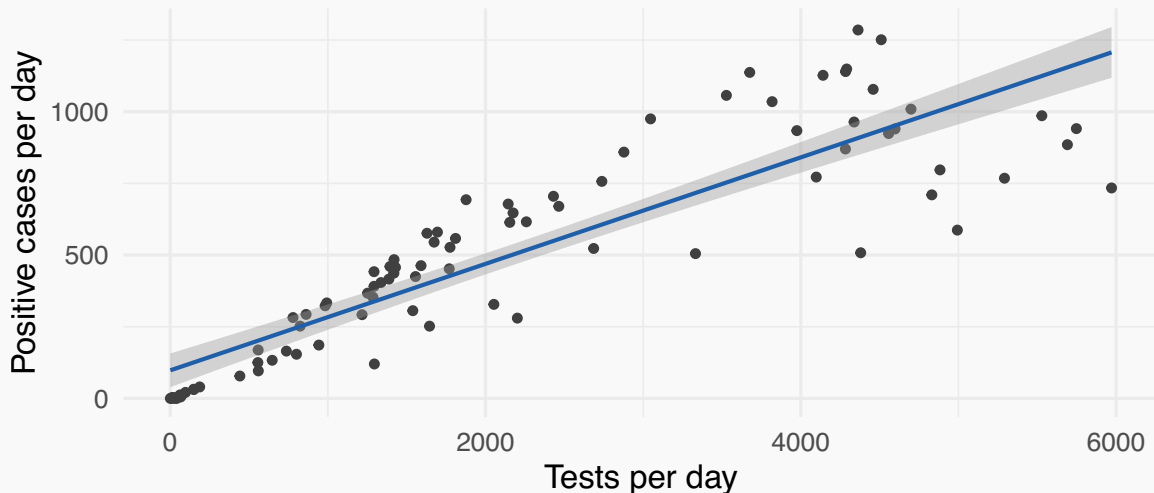
Daily counts for the city of Chicago in spring 2020, $n = 89$ days: the number of **tests** performed and the number of **positive cases** (file `COVID-19.csv`).

1. Draw a scatter plot of the data.
2. Find the correlation coefficient, and test whether the number of tests and the number of cases are linearly related.
3. Regress cases on tests: find the estimated intercept and slope, and the estimate of σ .
4. Give a 98% confidence interval for the slope.
5. Predict the number of cases on a day with 10, 100, 1000 and 5000 tests.
6. Give 95% confidence intervals for the mean number of cases at these values, and 95% prediction intervals.

1. Scatter plot

```
library(ggplot2)
Covid <- read.csv("COVID-19.csv")

ggplot(Covid, aes(Tests, Cases)) +
  geom_point(size = 0.9, colour = "grey25") +
  geom_smooth(method = "lm", colour = "#1F5FA9", linewidth = 0.6) +
  labs(x = "Tests per day", y = "Positive cases per day") +
  theme_minimal(base_size = 10)
```



What do you see? Is a straight line a reasonable summary of these points?

2. Correlation

```
cor.test(Covid$Tests, Covid$Cases)

##
## Pearson's product-moment correlation
##
## data: Covid$Tests and Covid$Cases
## t = 17.405, df = 87, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.8244944 0.9206675
## sample estimates:
##      cor
## 0.8814077
```

2. Correlation

```
cor.test(Covid$Tests, Covid$Cases)

##
##  Pearson's product-moment correlation
##
## data:  Covid$Tests and Covid$Cases
## t = 17.405, df = 87, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.8244944 0.9206675
## sample estimates:
##      cor
## 0.8814077
```

Keep the value $t = 17.405$ in mind: the regression on the next slide gives the slope exactly the same t value.

3. The fitted regression

```
fit <- lm(Cases ~ Tests, data = Covid)
summary(fit)
```

```
##
## Call:
## lm(formula = Cases ~ Tests, data = Covid)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -472.73  -99.45    1.19   114.42   376.91
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  97.77689   29.54996   3.309  0.00136 **
## Tests         0.18572    0.01067  17.405 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 173.8 on 87 degrees of freedom
## Multiple R-squared:  0.7769, Adjusted R-squared:  0.7743
## F-statistic: 302.9 on 1 and 87 DF, p-value: < 2.2e-16
```

Fitted line: $\hat{Y} = 97.8 + 0.186 X$

$$\hat{\sigma} = 173.8$$

Interpretation of $\hat{\beta}_1$: For ¹⁰⁰⁰ any additional test, we detect an additional ¹⁸⁶ 0.186 positive case on average

4. A 98% confidence interval for the slope

```
confint(fit, level = 0.98)
```

```
##              1 %              99 %  
## (Intercept) 27.7441770 167.8096011  
## Tests       0.1604334   0.2110128
```

By hand, with $t(0.99; 87) = 2.370$:

$$= 1 - \frac{(1 - \alpha)}{2}$$

$$\hat{\beta}_1 \pm t(0.99; 87) SE(\hat{\beta}_1) =$$

Interpretation

With 98% confidence, one more test gives
between 0.16 and 0.21 positives on average
additional

5 and 6. Predictions, and two kinds of interval

```
new.data <- data.frame(Tests = c(10, 100, 1000, 5000))  
predict(fit, new.data, interval = "confidence")
```

##		fit	lwr	upr
## 1		99.63412	41.06609	158.2022
## 2		116.34920	59.25869	173.4397
## 3		283.49999	239.32699	327.6730
## 4		1026.39241	956.00421	1096.7806

```
predict(fit, new.data, interval = "prediction")
```

##		fit	lwr	upr
## 1		99.63412	-250.66579	449.9340
## 2		116.34920	-233.70671	466.4051
## 3		283.49999	-64.68252	631.6825
## 4		1026.39241	673.92353	1378.8613

5 and 6. Predictions, and two kinds of interval

```
new.data <- data.frame(Tests = c(10, 100, 1000, 5000))  
predict(fit, new.data, interval = "confidence")
```

```
##           fit           lwr           upr  
## 1    99.63412    41.06609    158.2022  
## 2   116.34920    59.25869    173.4397  
## 3   283.49999   239.32699    327.6730  
## 4  1026.39241  956.00421   1096.7806
```

```
predict(fit, new.data, interval = "prediction")
```

```
##           fit           lwr           upr  
## 1    99.63412  -250.66579    449.9340  
## 2   116.34920  -233.70671    466.4051  
## 3   283.49999   -64.68252    631.6825  
## 4  1026.39241   673.92353   1378.8613
```

 The prediction intervals at 10, 100 and 1000 tests have **negative** lower limits: the model allows a day with a negative number of cases.

5 and 6. Predictions, and two kinds of interval

```
new.data <- data.frame(Tests = c(10, 100, 1000, 5000))  
predict(fit, new.data, interval = "confidence")
```

```
##           fit           lwr           upr  
## 1    99.63412    41.06609    158.2022  
## 2   116.34920    59.25869    173.4397  
## 3   283.49999   239.32699    327.6730  
## 4  1026.39241  956.00421   1096.7806
```

```
predict(fit, new.data, interval = "prediction")
```

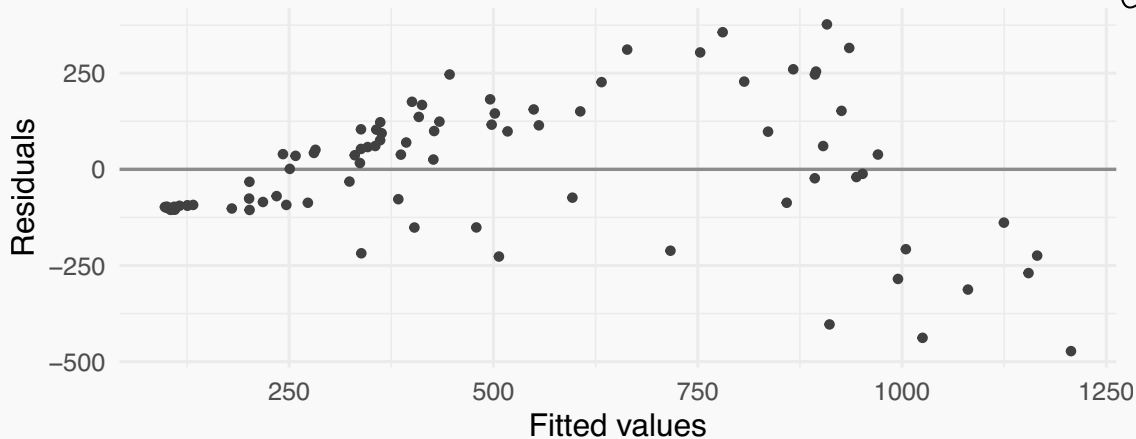
```
##           fit           lwr           upr  
## 1    99.63412  -250.66579    449.9340  
## 2   116.34920  -233.70671    466.4051  
## 3   283.49999   -64.68252    631.6825  
## 4  1026.39241   673.92353   1378.8613
```

 The prediction intervals at 10, 100 and 1000 tests have **negative** lower limits: the model allows a day with a negative number of cases.

The arithmetic is right: it is the model that is wrong. It gives the errors a normal distribution, which puts probability on negative values, while a number of cases can never be negative.

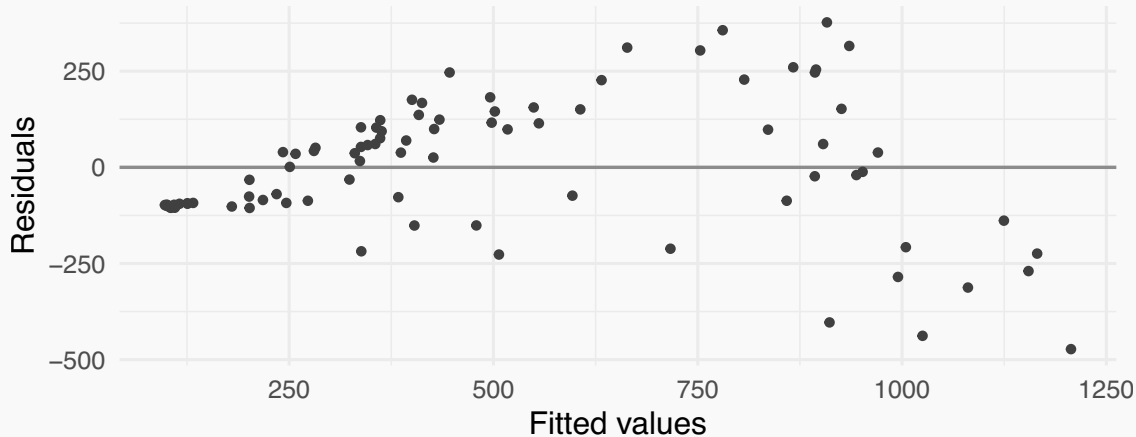
Look at the residuals

```
res <- data.frame(fitted = fitted(fit), resid = resid(fit))
ggplot(res, aes(fitted, resid)) +
  geom_hline(yintercept = 0, colour = "grey55") +
  geom_point(size = 0.9, colour = "grey25") +
  labs(x = "Fitted values", y = "Residuals") +
  theme_minimal(base_size = 10)
```



Look at the residuals

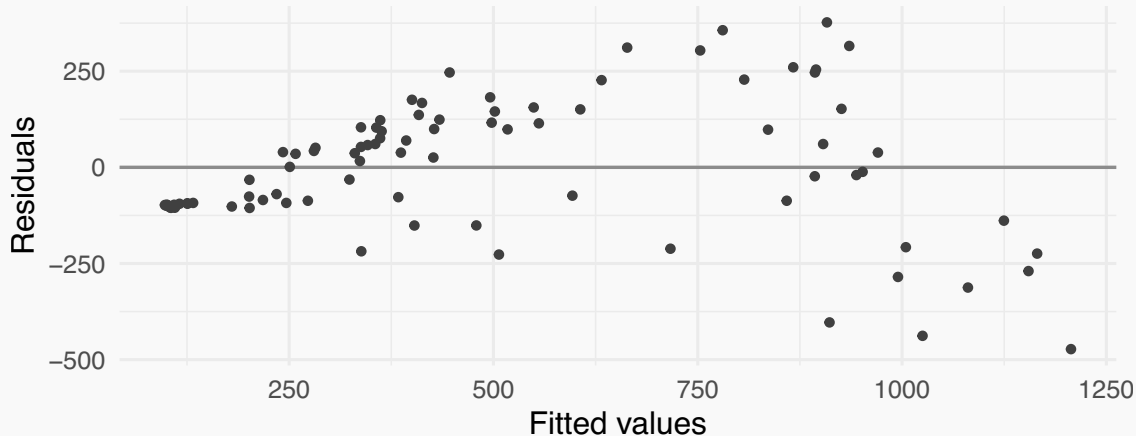
```
res <- data.frame(fitted = fitted(fit), resid = resid(fit))
ggplot(res, aes(fitted, resid)) +
  geom_hline(yintercept = 0, colour = "grey55") +
  geom_point(size = 0.9, colour = "grey25") +
  labs(x = "Fitted values", y = "Residuals") +
  theme_minimal(base_size = 10)
```



The spread of the residuals **grows** with the fitted value: their standard deviation is 54 on the third of days with fewest tests, and 257 on the third with most.

Look at the residuals

```
res <- data.frame(fitted = fitted(fit), resid = resid(fit))
ggplot(res, aes(fitted, resid)) +
  geom_hline(yintercept = 0, colour = "grey55") +
  geom_point(size = 0.9, colour = "grey25") +
  labs(x = "Fitted values", y = "Residuals") +
  theme_minimal(base_size = 10)
```

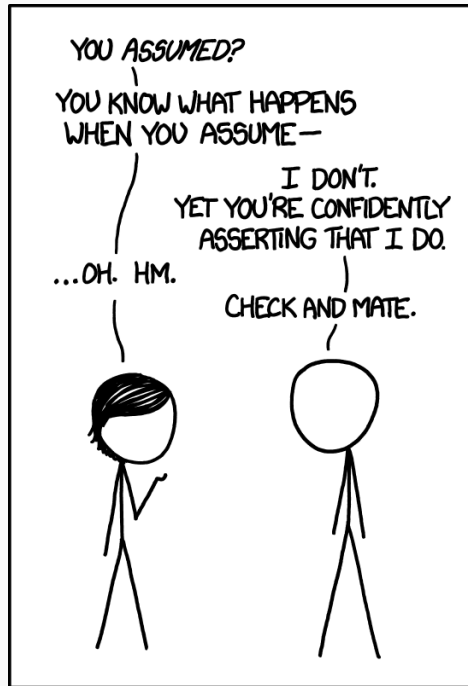


The spread of the residuals **grows** with the fitted value: their standard deviation is 54 on the third of days with fewest tests, and 257 on the third with most.

And the 89 observations are **consecutive days**: one day's error is likely to be close to the next day's.

Diagnostics and Remedial Measures

You assumed?



Every interval and test since Lecture 5 has rested on assumptions about the errors. Today we write them down, and start checking them.

xkcd.com/1339 (CC BY-NC 2.5)

Model assumptions

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i, \quad i = 1, \dots, n$$

Key assumptions of the simple linear regression model:

- (L) **Linearity**: the mean of Y is linear in X , $E(Y | X) = \beta_0 + \beta_1 X$
- (I) **Independence**: the errors are independent, or at least uncorrelated:
 $Cov(\epsilon_i, \epsilon_j) = 0$ for $i \neq j$
- (N) **Normality**: $\epsilon_i \sim N(0, \sigma^2)$
- (E) **Equal variance**: $Var(\epsilon_i) = \sigma^2$ for every i

Model assumptions

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i, \quad i = 1, \dots, n$$

Key assumptions of the simple linear regression model: $\Rightarrow E[\epsilon] = 0$

- (L) **Linearity**: the mean of Y is linear in X , $E(Y | X) = \beta_0 + \beta_1 X$
- (I) **Independence**: the errors are independent, or at least uncorrelated:
 $Cov(\epsilon_i, \epsilon_j) = 0$ for $i \neq j$
- (N) **Normality**: $\epsilon_i \sim N(0, \sigma^2)$
- (E) **Equal variance**: $Var(\epsilon_i) = \sigma^2$ for every i

Inference relies on the model being true. If an assumption fails, the estimates can be **biased**, and the standard errors, intervals and tests can be **wrong**.

What each result needs

Result	Assumptions used
$\hat{\beta}_0, \hat{\beta}_1$ unbiased	L
$Var(\hat{\beta}_1) = \sigma^2 / S_{xx}$, Gauss-Markov (BLUE)	L, I, E
t and F tests, confidence and prediction intervals	L, I, E, N

What each result needs

Result	Assumptions used
$\hat{\beta}_0, \hat{\beta}_1$ unbiased	L
$Var(\hat{\beta}_1) = \sigma^2 / S_{xx}$, Gauss-Markov (BLUE)	L, I, E
t and F tests, confidence and prediction intervals	L, I, E, N

- Normality is only needed at the last line: least squares itself never uses it.

What each result needs

Result	Assumptions used
$\hat{\beta}_0, \hat{\beta}_1$ unbiased	L
$Var(\hat{\beta}_1) = \sigma^2 / S_{xx}$, Gauss-Markov (BLUE)	L, I, E
t and F tests, confidence and prediction intervals	L, I, E, N

- Normality is only needed at the last line: least squares itself never uses it.
- Even there, it matters less than it seems for $\hat{\beta}_1$: $\hat{\beta}_1 = \beta_1 + \sum_i k_i \epsilon_i$ is a weighted **sum of n errors**, so by the central limit theorem it is close to normal for moderate n , whatever the distribution of the ϵ_i .

What each result needs

Result	Assumptions used
$\hat{\beta}_0, \hat{\beta}_1$ unbiased	L
$Var(\hat{\beta}_1) = \sigma^2 / S_{xx}$, Gauss-Markov (BLUE)	L, I, E
t and F tests, confidence and prediction intervals	L, I, E, N

- Normality is only needed at the last line: least squares itself never uses it.
- Even there, it matters less than it seems for $\hat{\beta}_1$: $\hat{\beta}_1 = \beta_1 + \sum_i k_i \epsilon_i$ is a weighted **sum of n errors**, so by the central limit theorem it is close to normal for moderate n , whatever the distribution of the ϵ_i .
- A prediction interval is about $Y_{n+1} = \beta_0 + \beta_1 x_{n+1} + \epsilon_{n+1}$: it involves **a single error** that is not averaged. Its distribution matters directly, however large n is.

What each result needs

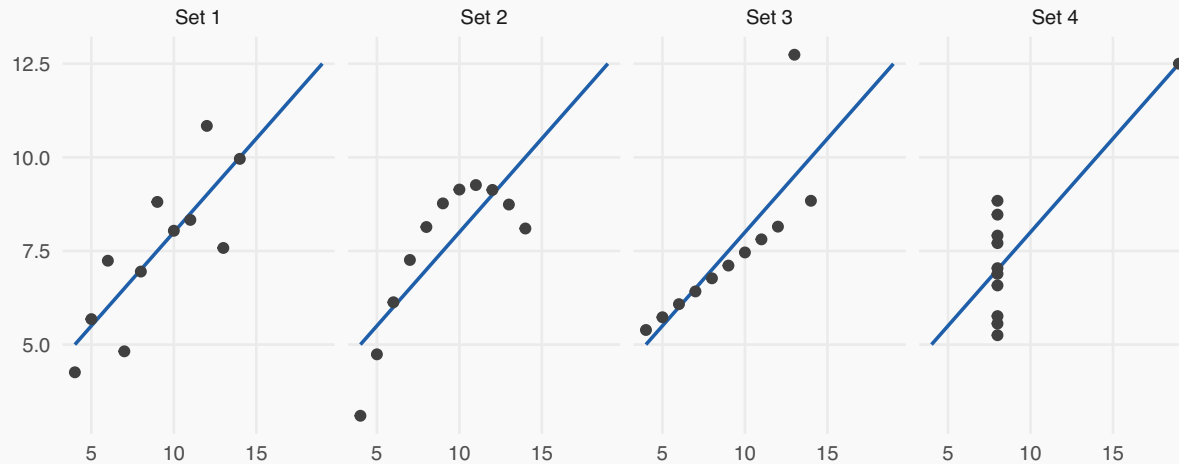
Result	Assumptions used
$\hat{\beta}_0, \hat{\beta}_1$ unbiased	L
$Var(\hat{\beta}_1) = \sigma^2 / S_{xx}$, Gauss-Markov (BLUE)	L, I, E
t and F tests, confidence and prediction intervals	L, I, E, N

- Normality is only needed at the last line: least squares itself never uses it.
- Even there, it matters less than it seems for $\hat{\beta}_1$: $\hat{\beta}_1 = \beta_1 + \sum_i k_i \epsilon_i$ is a weighted **sum of n errors**, so by the central limit theorem it is close to normal for moderate n , whatever the distribution of the ϵ_i .
- A prediction interval is about $Y_{n+1} = \beta_0 + \beta_1 x_{n+1} + \epsilon_{n+1}$: it involves **a single error** that is not averaged. Its distribution matters directly, however large n is.

We cannot observe the errors ϵ_i to perform statistical inference on them, so we check the assumptions with their approximations instead, the residuals $e_i = Y_i - \hat{Y}_i$, mostly by plotting them.

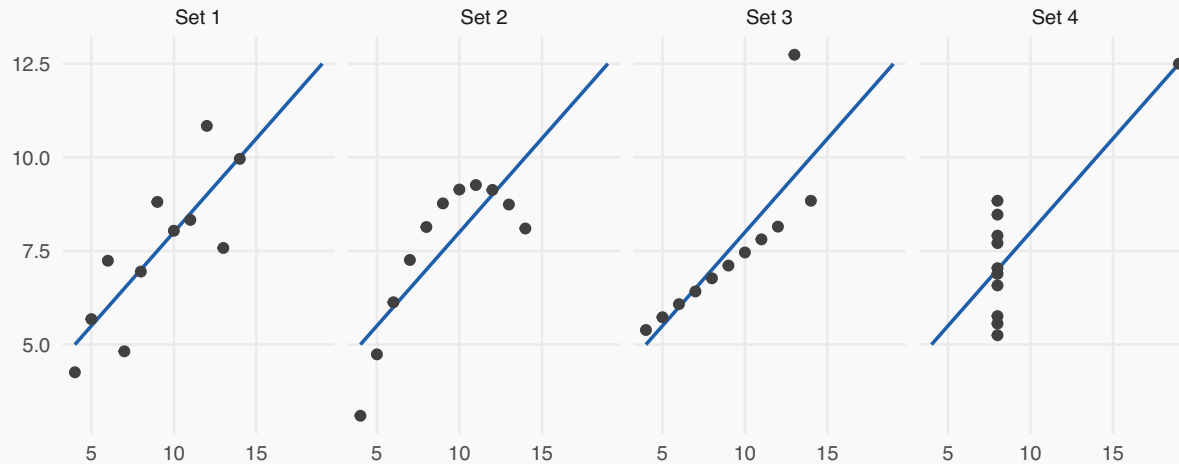
Checking assumptions: Anscombe's quartet

In Lecture 2: same r , different shapes. Anscombe's four datasets go further.



Checking assumptions: Anscombe's quartet

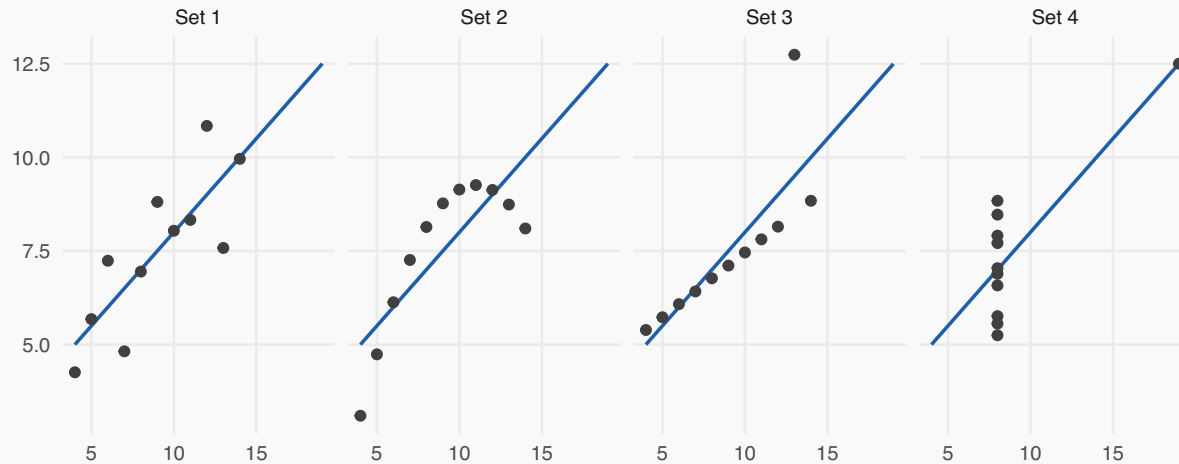
In Lecture 2: same r , different shapes. Anscombe's four datasets go further.



In all four: $\hat{\beta}_0 = 3.00$, $\hat{\beta}_1 = 0.500$, $r = 0.82$, $R^2 = 0.67$, $\hat{\sigma} = 1.24$. The same fitted line, the same fit.

Checking assumptions: Anscombe's quartet

In Lecture 2: same r , different shapes. Anscombe's four datasets go further.

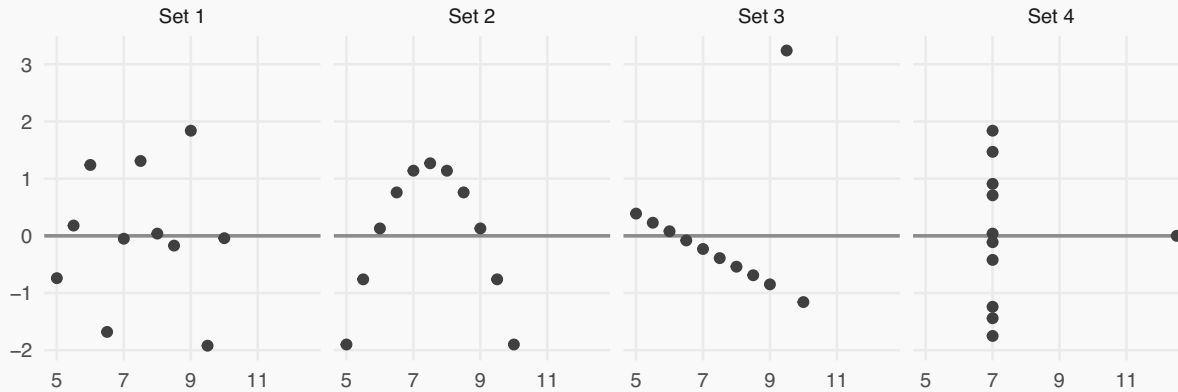


In all four: $\hat{\beta}_0 = 3.00$, $\hat{\beta}_1 = 0.500$, $r = 0.82$, $R^2 = 0.67$, $\hat{\sigma} = 1.24$. The same fitted line, the same fit.

The regression output cannot tell these four apart. A plot can.

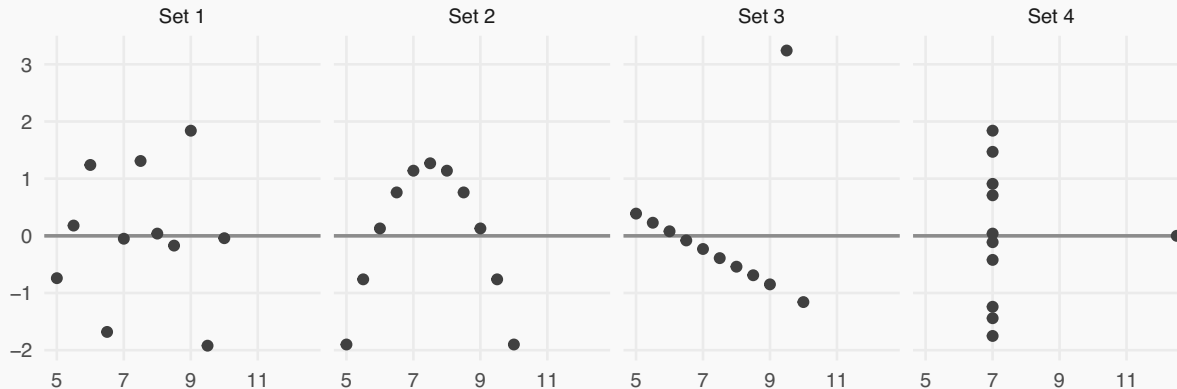
Residual plots

Residuals e_i plotted against the fitted values $\hat{Y}_i$:



Residual plots

Residuals e_i plotted against the fitted values $\hat{Y}_i$:



- Set 1: no pattern. This is what a residual plot should look like.
- Set 2: a clear curve. **Linearity** fails.
- Set 3: one outlier, while the other ten points lie exactly on a line.
- Set 4: one point alone determines the slope, yet its residual is zero.
⚠ A residual plot can miss a point with this much influence.

Which assumptions do the residuals put in doubt?

Which assumptions do the residuals put in doubt?

- (L) At 10 tests the fitted mean is 99.6 cases: ten times more positive cases than tests. A straight line with an intercept of 98 cannot be right near the origin.

Which assumptions do the residuals put in doubt?

- (L) At 10 tests the fitted mean is 99.6 cases: ten times more positive cases than tests. A straight line with an intercept of 98 cannot be right near the origin.
- (E) The spread grows with the number of tests. A single $\hat{\sigma}$ for every day is too large on quiet days, which is why the prediction intervals ran below zero.

Which assumptions do the residuals put in doubt?

- (L) At 10 tests the fitted mean is 99.6 cases: ten times more positive cases than tests. A straight line with an intercept of 98 cannot be right near the origin.
- (E) The spread grows with the number of tests. A single $\hat{\sigma}$ for every day is too large on quiet days, which is why the prediction intervals ran below zero.
- (I) Consecutive days resemble each other, so the errors are unlikely to be independent.

Which assumptions do the residuals put in doubt?

- (L) At 10 tests the fitted mean is 99.6 cases: ten times more positive cases than tests. A straight line with an intercept of 98 cannot be right near the origin.
- (E) The spread grows with the number of tests. A single $\hat{\sigma}$ for every day is too large on quiet days, which is why the prediction intervals ran below zero.
- (I) Consecutive days resemble each other, so the errors are unlikely to be independent.

Next lecture: a closer look at the residuals, formal checks of each assumption, and what to do when one fails.